

Units and Measurement

Physics Class 11 - Chapter 1

1.1 Introduction

Measurement is the process of comparing a physical quantity with a basic, arbitrarily chosen, internationally accepted reference standard called a unit. The result of a measurement is expressed by a numerical measure accompanied by a unit. While the number of physical quantities is large, we only need a limited number of units because they are interrelated. Units for base quantities are called fundamental or base units, and all others are derived units.

- Base units represent fundamental quantities that are independent of each other.
- Derived units are combinations of base units for other physical quantities.
- A system of units is a complete set of both base and derived units.
- Measurement involves comparing a quantity to an internationally accepted reference standard.

1.2 The International System of Units

Historically, various systems like CGS (centimetre, gram, second), FPS (foot, pound, second), and MKS (metre, kilogram, second) were used. Today, the internationally accepted system is the *Système International d'Unités*, or SI. This system, revised in 2018, is based on seven base units and is used globally in scientific, technical, and commercial work due to its convenient decimal system.

- SI is the International System of Units adopted for universal measurement usage.
- The system was developed by the BIPM and revised by the CGPM in 2018.
- It uses a decimal scheme, making unit conversions simple and convenient.
- SI is the standard for scientific, technical, industrial, and commercial work.

SI Base and Supplementary Units

The SI system is built upon seven base units: metre (m) for length, kilogram (kg) for mass, second (s) for time, ampere (A) for electric current, kelvin (K) for thermodynamic temperature, mole (mol) for amount of substance, and candela (cd) for luminous intensity. Additionally, there are two supplementary units defined for angles. The plane angle ($d\theta$) is the ratio of arc length (ds) to radius (r), while the solid angle ($d\Omega$) is the ratio of intercepted area (dA) to the square of the radius (r^2).

$$d\theta = ds/r \text{ radian}$$

$$d\Omega = dA/r^2 \text{ steradian}$$

- The seven base units are defined by fixed numerical values of universal constants.
- The radian (rad) is the unit for plane angles, which is a dimensionless quantity.
- The steradian (sr) is the unit for solid angles, also a dimensionless quantity.
- Modern definitions are revised to keep up with progress in measuring precision.

1.3 Significant Figures

Every measurement has an inherent uncertainty, and the reported result should indicate its precision. Significant figures include all reliably known digits plus the first uncertain digit. For instance, if a measurement is 1.62 s, '1' and '6' are certain, while '2' is uncertain. The number of significant figures does not change with a change of units, emphasizing that the precision depends on the measuring instrument's least count.

- Significant figures represent the digits known reliably plus the first uncertain digit.
- The number of significant digits indicates the precision of a measurement.
- Changing units (e.g., cm to m) does not alter the number of significant figures.
- A choice of units should not mislead regarding the precision of the measurement.

Rules for Determining Significant Figures

Specific rules help determine significant figures: 1. All non-zero digits are significant. 2. Zeros between non-zero digits are significant. 3. For numbers less than 1, leading zeros are not significant. 4. Trailing zeros in a number without a decimal are not significant (unless from measurement). 5. Trailing zeros in a number with a decimal point are significant. To avoid ambiguity, reporting measurements in scientific notation ($a \times 10^b$) is recommended.

- Non-zero digits and zeros between them are always significant.
- Leading zeros in decimals (like 0.00 in 0.0023) are never significant.
- Trailing zeros are significant only if the number contains a decimal point.
- Scientific notation clarifies the number of significant figures in trailing zeros.

1.3.1 Rules for Arithmetic Operations

Calculations involving measured values must reflect the uncertainty of the original data. In multiplication or division, the final result should have the same number of significant figures as the input value with the least significant figures. In addition or subtraction, the result should have the same number of decimal places as the input value with the least decimal places. This ensures that the result is not more precise than the original measurements.

- Multiplication/Division: Follow the least number of significant figures.
- Addition/Subtraction: Follow the least number of decimal places.
- Calculated density should match the precision of mass or volume measurements.
- These rules prevent results from appearing more accurate than the actual data.

1.3.2 Rounding off Uncertain Digits

Rounding is necessary when results contain more than one uncertain digit. If the digit to be dropped is greater than 5, the preceding digit is increased by 1; if less than 5, it remains unchanged. If the digit is exactly 5, the preceding digit is increased by 1 only if it is odd, and left unchanged if it is even. In complex multi-step calculations, it is best to retain one extra digit in intermediate steps to avoid cumulative rounding errors.

- Increase the preceding digit by 1 if the digit to be dropped is ≥ 5 .
- Leave the preceding digit unchanged if the digit to be dropped is < 5 .
- If dropping 5, the even-odd rule maintains the balance of rounding errors.
- Retaining extra digits in intermediate steps prevents additional errors.

1.4 Dimensions of Physical Quantities

Dimensions are the powers to which base quantities are raised to represent a physical quantity. There are seven base dimensions: [L], [M], [T], [A], [K], [cd], and [mol]. In mechanics, most quantities are written in terms of [M], [L], and [T]. For example, Volume has dimensions $[L^3]$, while Force, being mass times acceleration, has dimensions $[MLT^{-2}]$. Dimensional analysis focuses on the 'type' of quantity rather than its magnitude.

- Dimensions are expressed as base quantities raised to certain powers or exponents.
- Square brackets [] are used to indicate that we are dealing with dimensions.
- Physical quantities like speed and velocity share the same dimensions $[LT^{-1}]$.
- Quantities with zero dimensions in a base are marked with a power of 0.

1.5 Dimensional Formulae and Equations

A dimensional formula is an expression showing which base quantities represent the dimensions of a quantity. For example, $[M^0L^3T^0]$ is the dimensional formula for volume. A dimensional equation is formed by equating a physical quantity with its dimensional formula, such as $[F] = [MLT^{-2}]$. These equations are derived from the relationships between physical quantities and are essential for various scientific applications.

- Dimensional formulae show the composition of a quantity in terms of base units.
- Dimensional equations equate the symbol of a quantity to its dimensional expression.
- They can be obtained from the equations relating different physical quantities.
- Formulae for a wide variety of quantities are standardly used in physics.

1.6 Dimensional Analysis and Its Applications

Dimensional analysis is used to check the consistency of equations and deduce relations between quantities. The 'Principle of Homogeneity of Dimensions' states that only quantities with the same dimensions can be added or subtracted. While a dimensionally wrong equation is definitely incorrect, a dimensionally correct equation is not necessarily right, as dimensional analysis cannot determine dimensionless constants or distinguish between quantities with the same dimensions.

- Principle of Homogeneity: Every term in an equation must have the same dimensions.
- Dimensional consistency is a preliminary test for the correctness of an equation.
- Method of dimensions can derive relations by comparing powers of base quantities.
- Limitations include the inability to find dimensionless constants like 2π .

